

Form Number : _____

Paper Code : 3002CM370218011

English



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 – 2020)

NEET : PRACTICE PAPER

TARGET : PRE-MEDICAL - 2020

Test Type : PT # 05

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TEST DATE : 13-04-2020

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1. A seat marked with Reg. No. will be allotted to each student. The student should ensure that he/she occupies the correct seat only. If any student is found to have occupied the seat of another student, both the students shall be removed from the examination and shall have to accept any other penalty imposed upon them.
2. Duration of Test is **3 Hours** and Questions Paper Contains **180** Questions. The Max. Marks are **720**.
3. Student can not use log tables and calculators or any other material in the examination hall.
4. Student must abide by the instructions issued during the examination, by the invigilators or the centre incharge.
5. Before attempting the question paper ensure that it contains all the pages and that no question is missing.
6. Each correct answer carries 4 marks, while **1 mark will be deducted for every wrong answer**. Guessing of answer is harmful.
7. A candidate has to write his / her answers in the OMR sheet by darkening the appropriate bubble with the help of **Blue / Black Ball Point Pen only** as the correct answer(s) of the question attempted.
8. **Use of Pencil is strictly prohibited.**

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TOPIC COVER

PHYSICS : Ray Optics + Thermodynamics + Modern Physics + S.H.M. + Wave Optics

CHEMISTRY : Surface Chemistry + Metallurgy + Chemical Kinetics + p-Block, s-Block + Chemical Bonding

BIOLOGY : Animal Diversity + Nervous System + Cockroach

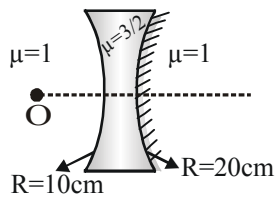
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. A thin glass (refractive Index 1.5) lens has optical power of $-5D$ in air. Its optical power in a liquid medium with refractive Index 1.6 will be :-

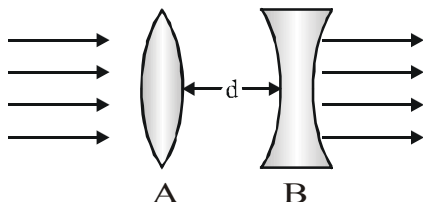
(1) $25 D$ (2) $\frac{5}{8} D$ (3) $1 D$ (4) $-1 D$

2. Find the power of given system :-



(1) $-10 D$ (2) $-15 D$
(3) $-20 D$ (4) $-25 D$

3. A convex lens A of focal length 20 cm and concave lens B of focal length 5 cm are kept along the same axis with distance d between them as shown in figure. If parallel beam is incident on lens A and emerge as parallel beam from B, then d is equal to (in cm) :-

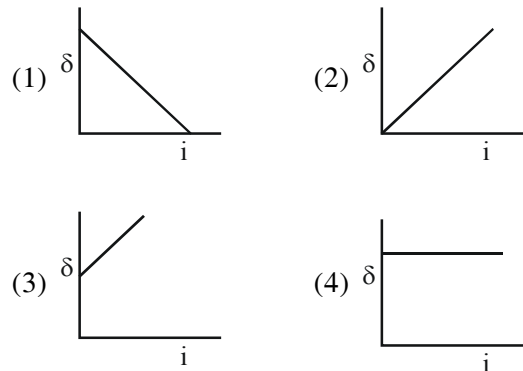


(1) 15 (2) 20
(3) 22.5 (4) 25

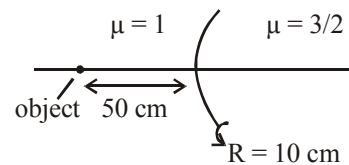
4. The distance between object and screen is D . Real images of an object are formed on screen for two position of lens separated by distance d . The ratio between the size of two images will be :-

(1) $\frac{D}{d}$ (2) $\frac{D^2}{d^2}$
(3) $\frac{(D-d)^2}{(D+d)^2}$ (4) $\sqrt{\frac{D}{d}}$

5. Graph between angle of deviation and incident angle in single reflection :-



- 6.



Find Image distance :-

(1) $+50 \text{ cm}$ (2) -50 cm
(3) $+10 \text{ cm}$ (4) -10 cm

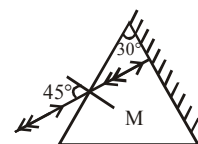
7. In convex lens of focal length F , the minimum distance between object and its real image :-

(1) $3F$ (2) $4F$ (3) 0 (4) $2F$

8. Two lens of focal length -20 cm and $+10 \text{ cm}$ are put combination, find power of combination :-

(1) $-1 D$ (2) $-2 D$ (3) $+5 D$ (4) $+2 D$

- 9.



In the given figure if light retrace its path, then refractive index of the prism is

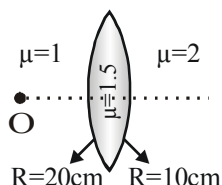
(1) $\frac{\sqrt{2}}{3}$ (2) $\sqrt{2}$ (3) $\sqrt{\frac{3}{2}}$ (4) $\frac{\sqrt{3}}{2}$

10. A lens is made of flint glass (refractive index is 1.5) when lens is immersed in a liquid of refractive index 1.25, the focal length becomes

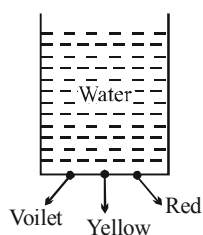
(1) 1.25 times (2) 2.5 times
(3) 0.5 times (4) 0.33 times

11. Find the power of given lens :-

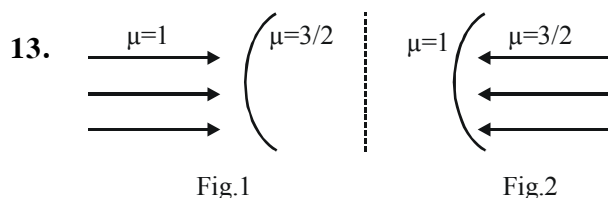
(1) +5 D
(2) +7.5 D
(3) +2.5 D
(4) -1.25 D



12. Which of following colour have more apparent depth viewed normally (In given diagram) :-



(1) Red (2) Violet
(3) Yellow (4) Same for all

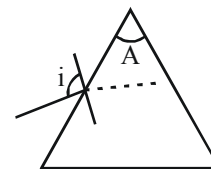


In given diagrams the surface will give rise to :-

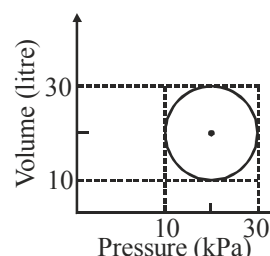
- (1) In first figure → Convergent beam
In second figure → Divergent beam
(2) In first figure → Convergent beam
In second figure → Convergent beam
(3) In first figure → Divergent beam
In second figure → Divergent beam
(4) In first figure → Divergent beam
In second figure → Convergent beam

14. Find the angle of minimum deviation, when angle of incidence is two times of angle of refraction in that surface and refractive index of prism is $\sqrt{3}$

(1) 30°
(2) 20°
(3) 40°
(4) 60°



15. The heat energy absorbed by a system in going through a cyclic process shown in figure is :-



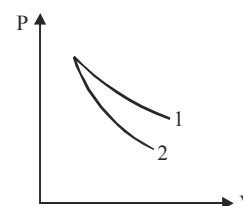
(1) $10^3 \pi$ J (2) $10^2 \pi$ J
(3) $10^4 \pi$ J (4) $10^7 \pi$ J

16. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is P_1 and final pressure is P_3 . The total work done is W . Then :-

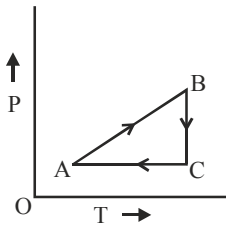
(1) $P_3 > P_1$, $W > 0$
(2) $P_3 < P_1$, $W < 0$
(3) $P_3 > P_1$, $W < 0$
(4) $P_3 = P_1$, $W = 0$

17. P-V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to :-

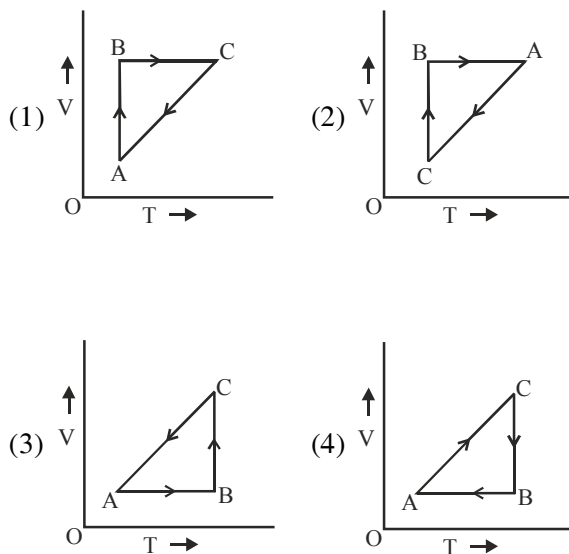
(1) He and O_2
(2) O_2 and He
(3) He and Ar
(4) O_2 and N_2



18. A cyclic process is shown in the P-T diagram. Which of the



curves show the same process on a V-T diagram:-



19. A gas is heated at constant pressure. The fraction of heat supplied used for external work is :-

- (1) $\frac{1}{\gamma}$ (2) $\left(1 - \frac{1}{\gamma}\right)$
(3) $\gamma - 1$ (4) $\left(1 - \frac{1}{\gamma^2}\right)$

20. At atmospheric pressure, 2g of water having a volume of 2.00 cm^3 becomes 3342 cm^3 of steam when boiled. The latent heat of vaporization of water is 539 cal/g at 1 atm. What is the amount of heat added to the system:-

- (1) 2156 cal (2) 1078 cal
(3) 539 cal (4) 269.5 cal

21. For free expansion of a gas which of the following is true :-

- (1) $Q = W = 0$ and $\Delta E_{\text{int}} = 0$
(2) $Q = 0$, $W > 0$ and $\Delta E_{\text{int}} = -W$
(3) $W = 0$, $Q > 0$ and $\Delta E_{\text{int}} = Q$
(4) $W = 0$, $Q < 0$ and $\Delta E_{\text{int}} = 0$

22. 80 gm of water at 30°C is poured on a large block of ice at 0°C . The mass of ice that melts is :-

- (1) 30 gm (2) 80 gm
(3) 150 gm (4) 1600 gm

23. Melting point of dry ice :-

- (1) Increases with increasing pressure
(2) Decreases with increasing pressure
(3) Is independent of pressure
(4) Is proportional to pressure

24. The temperature on Fahrenheit thermometer at which the reading of a Fahrenheit thermometer will be double that of Centigrade thermometer is :-

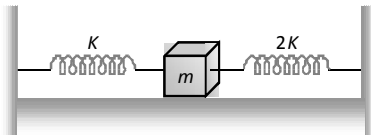
- (1) 160°C (2) 160 F
(3) 320 F (4) 320°C

25. The ratio of the de Broglie wavelengths of an electron of energy 10 eV to that of a person of mass 66 kg travelling at a speed of 100 km/hour is of the order of :

- (1) 10^{34} (2) 10^{27} (3) 10^{17} (4) 10^{-10}

26. One milliwatt of light of wavelength $4560 \text{ \AA}$ is incident on a caesium surface of work function 1.9 eV. Given that quantum efficiency of photoelectric emission is 0.5%, Planck constant $h = 6.62 \times 10^{-34} \text{ J-sec}$, velocity of light $C = 3 \times 10^8 \text{ m/s}$, the photoelectric current liberated is :

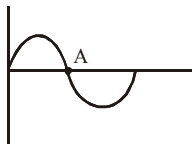
- (1) $1.856 \times 10^{-6} \text{ amp}$
(2) $1.856 \times 10^{-7} \text{ amp}$
(3) $1.856 \times 10^{-5} \text{ amp}$
(4) $1.856 \times 10^{-4} \text{ amp}$

27. When a U^{238} nucleus originally at rest, decays by emitting an alpha-particle having a speed 'u', the recoil speed of the residual nucleus is :
- (1) $-\frac{4u}{238}$ (2) $\frac{4u}{238}$ (3) $-\frac{4u}{234}$ (4) $\frac{4u}{234}$
28. Ionisation energy of a hydrogen like atom is 16 Rydberg. Difference of potential energy of an electron in this atom in 2nd state and 3rd excited state is
- (1) 27.2 eV (2) 40.8 eV
(3) 54.4 eV (4) 81.6 eV
29. Wavelength of K_{α} , K_{β} and L_{α} X-rays for an element are λ_1 , λ_2 , λ_3 respectively then
- (1) $\lambda_1 < \lambda_2 < \lambda_3$
(2) $\lambda_1 > \lambda_2 > \lambda_3$
(3) $\lambda_2 < \lambda_1 < \lambda_3$
(4) $\lambda_3 < \lambda_1 < \lambda_2$
30. A source of 120 watt having luminous efficiency 25% emits light of wavelength $6600\text{\AA}$ incident on a metal of quantum efficiency 50% in a photoelectric experiment. Value of saturation current is
- (1) 4.6 A (2) 12.8 A
(3) 8.0 A (4) 16.5 A
31. Ratio of de-Broglie's wavelength associated with a Bohr's electron in 2nd state of Li^{++} and 3rd state of He^{+} is
- (1) 3 : 1 (2) 4 : 9
(3) 9 : 4 (4) 1 : 3
32. Two springs of force constants K and $2K$ are connected to a mass as shown below. The frequency of oscillation of the mass is
- 
- (1) $(1/2\pi)\sqrt{(K/m)}$
(2) $(1/2\pi)\sqrt{(2K/m)}$
(3) $(1/2\pi)\sqrt{(3K/m)}$
(4) $(1/2\pi)\sqrt{(m/K)}$
33. The S.H.M. of a particle is given by the equation $y = 3 \sin \omega t + 4 \cos \omega t$. The amplitude is
- (1) 7 (2) 1 (3) 5 (4) 12
34. The time period of a simple pendulum in a lift descending with constant acceleration g is
- (1) $T = 2\pi\sqrt{\frac{l}{g}}$
(2) $T = 2\pi\sqrt{\frac{l}{2g}}$
(3) Zero
(4) Infinite
35. In simple harmonic motion, the ratio of acceleration of the particle to its displacement at any time is a measure of
- (1) Spring constant
(2) Angular frequency
(3) (Angular frequency)²
(4) Restoring force
36. A simple pendulum is made of a body which is a hollow sphere containing mercury suspended by means of a wire. If a little mercury is drained off, the period of pendulum will
- (1) Remains unchanged
(2) Increase
(3) Decrease
(4) Become erratic

37. A wave travelling along the x-axis is described by the equation $y(x, t) = 0.005 \cos(\alpha x - \beta t)$. If the wavelength and the time period of the wave is 0.08m and 2.0 s respectively then α and β in appropriate units are

- (1) $\alpha = 25.00\pi$, $\beta = \pi$
 (2) $\alpha = \frac{0.08}{\pi}$, $\beta = \frac{2.0}{\pi}$
 (3) $\alpha = \frac{0.04}{\pi}$, $\beta = \frac{1.0}{\pi}$
 (4) $\alpha = 12.50\pi$, $\beta = \frac{\pi}{2.0}$

38. For the given wave equation $y = 10\sin(-2\pi t - 4\pi x)$ what will be correct for velocity of particle A.



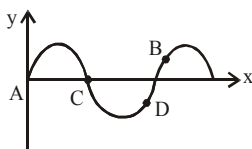
- (1) 20π downwards (2) 0.5 downwards
 (3) 0.5 upwards (4) 20π upwards

39. The equation of wave is given by :

$y = 10\sin\left(\frac{2\pi t}{30} + \alpha\right)$. If the displacement is 5 cm at $t = 0$, then the total phase of $t = 7.5$ sec will be

- (1) $\frac{\pi}{3}$ rad (2) $\frac{2\pi}{5}$ rad
 (3) $\frac{2\pi}{3}$ rad (4) $\frac{\pi}{2}$ rad

40. For a given transverse wave equation $y = 10 \sin(-10\pi t - 5\pi x)$ find out the direction of following points A, B, C, D in figure.



- (1) $\uparrow, \downarrow, \downarrow, \uparrow$ (2) $\downarrow, \uparrow, \downarrow, \uparrow$
 (3) $\uparrow, \uparrow, \downarrow, \uparrow$ (4) $\downarrow, \uparrow, \downarrow, \downarrow$

41. White light is used to illuminate the two slits in Young's experiment separation between the slits is b and the screen is at a distance d ($\gg b$) from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. The missing wavelengths are

- (1) $\lambda = \frac{b^2}{d}$ (2) $\lambda = \frac{b^2}{5d}$
 (3) $\lambda = \frac{b^2}{3d}$ (4) All of these

42. In a Young's double-slit interference experiment the fringe pattern is observed on a screen placed at a distance D . The slit are separated by d and are illuminated by light of wavelength λ . The distance from the central point where the intensity falls to half the maximum is

- (1) $\frac{\lambda D}{3d}$ (2) $\frac{\lambda D}{2d}$ (3) $\frac{\lambda D}{d}$ (4) $\frac{\lambda D}{4d}$

43. The maximum intensity in Young's double-slit experiment is I_0 . Distance between the slits is $d = 5\lambda$, where λ is the wavelength of monochromatic light used in the experiment. What will be the intensity of light in front of one of the slits on a screen at a distance $D = 10d$?

- (1) $\frac{I_0}{2}$ (2) $\frac{3}{4}I_0$ (3) I_0 (4) $\frac{I_0}{4}$

44. Young's double slit experiment is made in a liquid. The 10th bright fringe in liquid lies where 6th dark fringe lies in vacuum. The refractive index of the liquid is approximately
- (1) 1.8 (2) 1.54 (3) 1.67 (4) 1.2

45. In a YDSE experiment, I_0 is given to be the intensity of the central bright fringe and β is the fringe width. Then, at a distance y from central Bright Fringe, the intensity will be

- (1) $I_0 \cos(\pi y/\beta)$ (2) $I_0 \cos^2(\pi y/\beta)$
 (3) $I_0 \cos(2\pi y/\beta)$ (4) $I_0 \cos^2(\pi y/2\beta)$

46. A colloidal solution is subjected to an electrical field. The particles move towards anode. The coagulation of the same solution is studied using NaCl, BaCl₂ and AlCl₃ solution. Their coagulating power of electrolyte should be in which of the following order:
- NaCl > BaCl₂ > AlCl₃
 - BaCl₂ > AlCl₃ > NaCl
 - AlCl₃ > BaCl₂ > NaCl
 - BaCl₂ > NaCl > AlCl₃
47. Which of the following is most effective in coagulating a ferric hydroxide sol
- KCl
 - KNO₃
 - K₂SO₄
 - K₃[Fe(CN)₆]
48. On addition of one ml solution of 10% NaCl to 10 ml gold sol in the presence of 0.0250 gm of starch the coagulation is just prevented. Starch has the following gold number
- 0.025
 - 0.25
 - 2.5
 - 25
49. The gold numbers of four protective colloids O, P, Q and R are 0.005, 0.01, 0.1 and 0.5 respectively. The decreasing order of their protective power is -
- R, Q, P, O
 - O, P, Q, R
 - P, Q, R, O
 - Q, R, O, P
50. Which of the following statements is incorrect regarding physisorptions ?
- More easily liquefiable gases are adsorbed readily
 - Under high pressure it results into multi molecular layer on adsorbent surface
 - Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive
 - It occurs because of van der Waal's forces
51. Which of the following metals is protected by a layer of its own oxide—
- Aluminium
 - Silver
 - Gold
 - Iron
52. Cupellation process is involved in the metallurgy of—
- Copper
 - Silver
 - Gold
 - Lead
53. The chemical processes in the production of steel from haematite ore involve
- Reduction
 - Oxidation
 - Reduction followed by oxidation
 - Oxidation followed by reduction
54. Which of the following is used in the extractive metallurgy of magnesium -
- Fused salt electrolysis
 - Self reduction
 - Aqueous solution of electrolytes
 - Thermite reduction
55. The major role of fluorspar (CaF₂), which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na₃AlF₆), is—
- As a catalyst.
 - To make the fused mixture very conducting
 - To lower the temperature of the melt
 - To decrease the rate of oxidation of carbon at the anode
56. Tin & lead can be refined by-
- Cupellation
 - Liquation
 - Poling
 - Bessemerisation

57. If the rate constant of a reaction is $2 \times 10^{-3} \text{ s}^{-1}$. What is the order of the reaction :-

- (1) 0 (2) 1
(3) 2 (4) 3

58. The plot of $\log k$ vs $1/T$ yields a straight line. The slope of the line would be equal to :-

- (1) $-\frac{E_a}{R}$ (2) $-\frac{E_a}{2.303R}$
(3) $\frac{E_a}{R}$ (4) $\frac{E_a}{2.303R}$

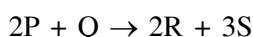
59. $A \rightarrow \text{product}$ is a first order reaction. After three 'half lives' what fraction of A would be left ?

- (1) 6.25% (2) 12.5%
(3) 50% (4) 75%

60. For every 10°C rise of temperature the rate is doubled then if the temperature is increased from 10°C to 100°C then rate becomes :-

- (1) 400 times
(2) 512 times
(3) 112 times
(4) 614 times

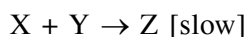
61. The rate of disappearance of Q in the reaction :-



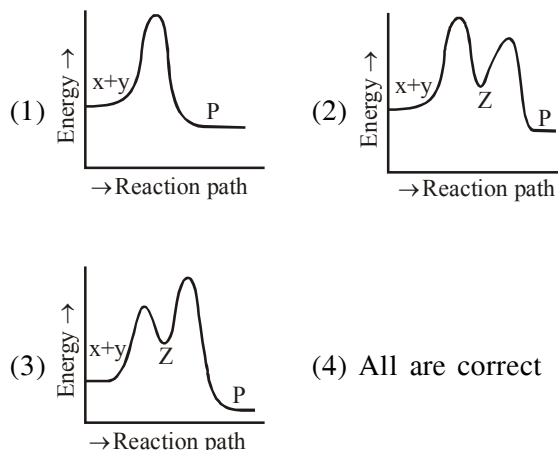
is $2 \times 10^{-2} \text{ moles } \ell^{-1}\text{s}^{-1}$. Which of the following statement is not true :-

- (1) $-\frac{d[P]}{dt} = 4 \times 10^{-2} \text{ moles } \ell^{-1}\text{s}^{-1}$
(2) $+\frac{d[S]}{dt} = 6 \times 10^{-2} \text{ moles } \ell^{-1}\text{s}^{-1}$
(3) $+\frac{1}{2} \cdot \frac{d[R]}{dt} = 2 \times 10^{-2} \text{ moles } \ell^{-1}\text{s}^{-1}$
(4) $+\frac{d[R]}{dt} = 2 \times 10^{-2} \text{ moles } \ell^{-1}\text{s}^{-1}$

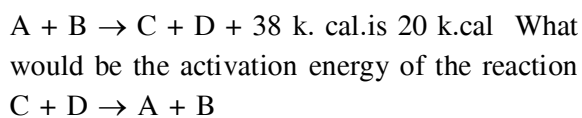
62. Certain exothermic reaction takes place in two steps



The energy level diagram corresponding to the reaction is :-



63. The activation energy of the reaction :



- (1) 20 k. cal (2) -20 k. cal
(3) 18 k. cal (4) 58 k. cal

64. Following data are obtained for the reaction $A + B \rightarrow \text{product}$:-

[A]	[B]	Rate
(mol ℓ^{-1})	(mol ℓ^{-1})	(mol $\ell^{-1} \text{ s}^{-1}$)
4×10^{-2}	4×10^{-2}	4×10^{-2}
4×10^{-2}	8×10^{-2}	4×10^{-2}
2×10^{-2}	8×10^{-2}	2×10^{-2}

The reaction confirms the rate law :-

- (1) $-\frac{d[B]}{dt} = k[A]$ (2) $-\frac{d[B]}{dt} = k[A][B]^2$
(3) $-\frac{d[B]}{dt} = k[A][B]$ (4) $-\frac{d[A]}{dt} = k[B]$

65. Arrange P_4O_6 , P_4O_8 , P_4O_{10} in increasing order of sum of σ & π Bonds present

- (1) $P_4O_8 < P_4O_{10} < P_4O_6$
(2) $P_4O_6 < P_4O_8 < P_4O_{10}$
(3) $P_4O_{10} < P_4O_6 < P_4O_8$
(4) All are having same number of bonds.

66. Which allotropic form of sulphur is soluble in CS_2 .
 (1) Rhombic sulphur
 (2) Monoclinic sulphur
 (3) Both forms
 (4) None
67. Hybridisation of phosphorus when PCl_5 exists in solid form
 (1) sp^3 , sp^3d (2) sp^3 , sp^3d^2
 (3) sp^3 , sp^3 (4) sp^3d , sp^3d^2
68. Which of the following statement is/are correct in context of white phosphorous.
 (i) White phosphorus is a translucent white waxy solid
 (ii) It is poisonous
 (iii) On reaction with NaOH it disproportionates.
 (iv) It consists of discrete tetrahedral P_4 units.
 (v) It is more reactive in comparison to red and Black phosphorous.
 (1) (i), (ii), (iii), (iv)
 (2) (i), (ii), (iii), (v)
 (3) (i), (iii), (v), (iv)
 (4) (i), (ii), (iii), (iv), (v)
69. In Brown ring test complex formed is
 (1) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{+3}$ (2) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{+2}$
 (3) $[\text{Fe}(\text{H}_2\text{O})\text{NO}_3]^{+2}$ (4) $[\text{Fe}(\text{H}_2\text{O})_4(\text{NO}_2)]^{+2}$
70. Which of the following reactions gives same acidic product.
 (i) $\text{NO}_2 + \text{H}_2\text{O}$ (ii) $\text{N}_2\text{O}_3 + \text{H}_2\text{O}$
 (iii) $\text{N}_2\text{O}_5 + \text{H}_2\text{O}$ (iv) $\text{N}_2\text{O} + \text{H}_2\text{O}$
 (1) (i), (ii) (2) (ii), (iv)
 (3) (i), (iii) (4) (i), (ii), (iii)
71. $\text{NaOH} + \text{Cl}_2 \longrightarrow \text{A} + \text{B} + \text{H}_2\text{O}$
 $\downarrow \text{conc. H}_2\text{SO}_4$
 C
 C is a monobasic acid
 (1) A-NaOCl B-NaCl C-HOCl
 (2) A-NaOCl₃ B-NaCl C-HOCl
 (3) A-NaCl B-NaOCl₃ C-HCl
 (4) A-NaCl B-NaOCl₃ C-HOCl
72. Which of following is correct
 (1) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$: Bond angle
 (2) $\text{P} > \text{N} > \text{As} > \text{Sb}$: ΔH_{eg}
 (3) Hybridisation of sulphur in SF_4 is sp^3d and having 2 lone pair on sulphur.
 (4) ICl is more reactive than I_2 .
73. Bismuth does not form stable pentahalide because of -
 (1) Its higher electronegativity
 (2) Its smaller size
 (3) Inert pair effect
 (4) Non availability of d-orbitals
74. What is false about N_2O_5 ?
 (1) It is anhydride of HNO_3
 (2) It is a powerful oxidizing agent
 (3) Solid N_2O_5 is called nitronium nitrate
 (4) Structure of N_2O_5 contains no $[\text{N} \rightarrow \text{O}]$ bond
75. Vegetable colouring matter in presence of moisture is bleached by SO_2 due to -
 (1) Oxidation (2) Reduction
 (3) Sulphonation (4) Unsaturation
76. Of the following statements -
 (a) Cl_2 gas is dried by using conc. H_2SO_4
 (b) Fluorine has highest oxidising power
 (c) Oxidising power of halogens follows the order $\text{I}_2 > \text{Br}_2 > \text{Cl}_2$
 (d) HI is the strongest acid among HI , HBr , HCl
 (1) a, b and d are corrects
 (2) a, c are corrects
 (3) b, c are corrects
 (4) c, d are corrects
77. The property of halogen acids, that indicated incorrect is -
 (1) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$ thermal stability
 (2) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ reducing strength
 (3) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$ bond length
 (4) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$ acidic strength

78. Which among the following does not contains coordinate bond :-
 (a) NH_4^+ (b) CO_3^{2-} (c) NO_2^- (d) H_3O^+
 (1) All (2) b, c (3) c, d (4) a, d
79. Which one of the following compound has bond angle nearly 90° :-
 (1) NH_3 (2) H_2S (3) H_2O (4) CH_4
80. Which one of the following on heating will not give CO_2
 (1) CaCO_3 (2) Na_2CO_3
 (3) PbCO_3 (4) Li_2CO_3
81. Which of the following alkali metal ions has lowest ionic mobility in aqueous solution
 (1) Rb^+ (2) Cs^+ (3) Li^+ (4) Na^+
82. A solid compound 'X' on heating gives CO_2 gas and a residue. The residue mixed with water forms 'Y'. On passing an excess of CO_2 through 'Y' in water, a clear solution, 'Z' is obtained. On boiling 'Z', compound 'X' is reformed. The compound 'X' is
 (1) Na_2CO_3 (2) K_2CO_3
 (3) $\text{Ca}(\text{HCO}_3)_2$ (4) CaCO_3
83. Sodium carbonate is manufactured by Solvay process, the products that are recycled are
 (1) CO_2 and NH_3
 (2) CO_2 and NH_4Cl
 (3) NaCl , CaO
 (4) CaCl_2 , CaO
84. Which of the following metal is obtained by leaching process using a solution of NaCN and then precipitating the metal by addition of zinc dust –
 (1) Copper (2) Silver
 (3) Nickel (4) Iron
85. Mac-Aurthur forrest process is used for the extraction of –
 (1) Zn (2) Cu (3) Fe (4) Ag
86. Match **List-I** (Metal) with **List-II** (Process of Extraction) and select the correct answer using the codes given below the lists -
- | List - I | List - II |
|-----------------|-------------------------|
| (Metal) | (Process of Extraction) |
| A. Aluminium | a. Blast furnace |
| B. Iron | b. Mond process |
| C. Nickel | c. Bayer process |
| D. Copper | d. Cyanide process |
| | e. Froath floatation |
- Code:**
- | | A | B | C | D |
|-----|---|---|---|---|
| (1) | b | e | d | a |
| (2) | c | a | b | e |
| (3) | b | a | d | e |
| (4) | c | e | b | a |
87. Refining of silver is done by–
 (1) Liquation (2) Poling
 (3) Cupellation (4) Van arkel method
88. KF combines with HF to form KHF_2 . The compound contains the species
 (1) K^+ , F^- and H^+ (2) K^+ , F^- and HF
 (3) K^+ and $[\text{HF}_2]^-$ (4) $[\text{KHF}]^+$ and F^-
89. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which one of the following does not occur
 (1) Blue coloured solution is obtained
 (2) Na^+ ions are formed in the solution
 (3) Liquid NH_3 becomes good conductor of electricity
 (4) Liquid ammonia remains diamagnetic
90. Number of sigma & pi bonds in N_2O_5 is :-
 (1) 6, 0 (2) 2, 6 (3) 6, 2 (4) 5, 1

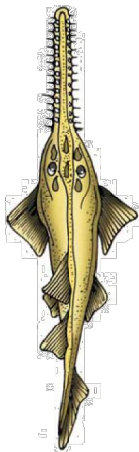
91. How many animals are marine from given below :
 Aurelia, Sycon, *Planaria*, Limulus, Octopus, Ophiura, Myxine, Pristis, Exocoetus
 (1) Eight (2) Seven (3) Five (4) Four
92. Which of the following is known as typical chordate ?
 (1) Ascidia (2) Amphioxus
 (3) Lamprey (4) Trygon
93. Which animals has similar larva to bipinnaria of echinodermites ?
 (1) Branchiostoma (2) Doliolum
 (3) Saccoglossus (4) Pila
94. Which of the following function is not performed by water vascular system ?
 (1) Transport of food (2) Locomotion
 (3) Reproduction (4) Respiration
95. In which of the following notochord is absent ?
 (1) larva of Branchiostoma
 (2) larva of Ascidia
 (3) Adult Salpa
 (4) Adult Amphioxus
96. Animal has true calcareous endoskeleton
 (1) Sea lily (2) Balanoglossus
 (3) Ascidia (4) Lancelet
97. Which of the following is present in all molluscans without any exception ?
 (1) Radula (2) Visceral hump
 (3) Unsegmented body (4) Shell
98. Choose correctly matched option :
 (1) Neries – Jointed appendages
 (2) Loligo – Radula for feeding
 (3) Ancylostoma – flame cells of excretion
 (4) Pleurobranchia – comb plates of locomotion
99. (a) Cnidoblasts are present for anchorage and defence.
 (b) Exhibit two body form polyp and medusa.
 (c) Show metagenesis.
 (d) Skeleton composed of calcium carbonate.
 How many statements are true for Adamsia ?
 (1) One (2) Two (3) Three (4) Four
100. Which of the following is not true for Euspongia ?
 (1) Water vascular system helpful in food gathering, respiratory exchange and removal of waste.
 (2) Sexes are not separate.
 (3) Choanocytes line the spongocoel and the canals.
 (4) Multicellular animals with cellular level of body organisation.
101. Which has a sucking circular mouth without jaw ?
 (1) Trygon (2) Myxine
 (3) Clarias (4) Ichthyophis
102. How many vertebrates found in marine water ?
 Ctenoplana, Amphioxus, Trygon, Carcharodon, Exocoetus, Echinus, Octopus, Aplysia
 (1) Three (2) Four (3) Five (4) Six
103. Which animal has incomplete digestive tract ?
 (1) Ancylostoma (2) Neries
 (3) Taenia (4) Planaria
104. Which of the following is not a dioecious ?
 (1) Neries (2) Wuchereria
 (3) Culex (4) Hirudinaria
105. Which of the following is not true about Balanoglossus ?
 (1) Circulatory system is of open type.
 (2) Excretory organ is proboscis gland.
 (3) Sexes are not separate.
 (4) Respiration takes place through gills.
106. Notochord absent in
 (1) Larva of Ascidia
 (2) Larva of Balanoglossus
 (3) Adult Ascidia
 (4) Both (2) and (3)

107. Choose an incorrect statement about the phylum of given animals :



- (1) Body is covered by a calcareous shell.
- (2) The mouth contains a file like rasping organ of feeding, called radula.
- (3) The gills have respiratory and excretory functions.
- (4) They are usually monoecious and oviparous.

108. In given figure Animal-A differs from Animal-B in lacking of



Animal - A



Animal - B

- (a) Two chambered heart
 - (b) Air bladder
 - (c) Operculum
 - (d) Notochord Persistent
 - (e) Placoid scale
 - (f) Ctenoid Scale
 - (g) Copulatory organ in male
- (1) (b), (c) and (f)
 - (2) (d), (e) and (g)
 - (3) (e) and (g)
 - (4) (d), (f) and (g)

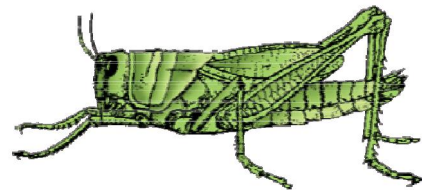
109. Choose a mismatched option :

- (1) Balanoglossus – Excretion by proboscis glands
- (2) Echinus – Excretion by water canal system
- (3) Doliolum – Notochord absent in adult
- (4) Pterophyllum – Oviparous and direct development

110. Which of the following not true about chordates ?

- (1) Central nervous system is dorsal, hollow and single.
- (2) Pharynx perforated by gill slits.
- (3) Myogenic pulsatile dorsal heart.
- (4) Solid, rod like notochord present ventral to nerve cord.

111. Select correct statements about given diagram.

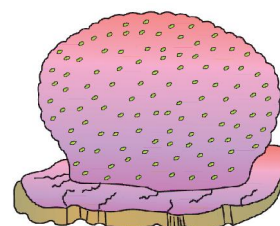


- (a) One pair antennae are present
 - (b) Excretion take place by malpighian tubules.
 - (c) Body covered by chitinous exoskeleton.
 - (d) Respiration take place by gills.
- (1) Two (2) Three (3) Four (4) One

112. Choose an incorrectly matched option :

- (1) Aplysia – Spirally coiled shell
- (2) Antedon – Water vascular system
- (3) Wuchereria – viviparous round worm
- (4) Ascidia – sexes are separate

113. How many statements are true about given animal ?



- (a) They have a water transport system.
(b) Their skeleton formed by spongin fibre.
(c) Choanocytes lines the spongocoel and the canals.
(d) Sexes are separate.
(1) Two (2) Three (3) One (4) Four

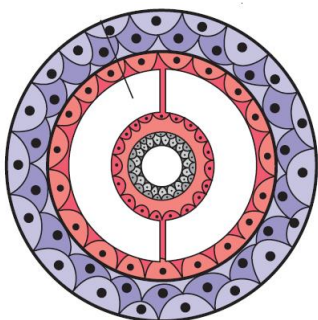
114. Metameric segmentation absent in

- (1) Laccifer (2) Ascidia
(3) Petromyzon (4) Balanoglossus

115. Which animals has paired jointed appendages ?

- (1) Salamandra (2) Pristis
(3) Culex (4) Neries

116. Choose an odd one about given diagram :



- (1) Aedes (2) Octopus
(3) Ancylostoma (4) Pheretima

117. The ___(i)___ neural system relays impulses from CNS to involuntary organs and ___(ii)___ relays from CNS to skeletal muscles?

- | (i) | (ii) |
|---------------------|-------------------------------|
| (1) Somatic | Autonomus neural system |
| (2) Sympathetic | Parasympathetic neural system |
| (3) Parasympathetic | Autonomus neural system |
| (4) Autonomous | somatic neural system |

118. Read the following statements (A-D)

- (A) In our body the neural system and the endocrine system jointly coordinate and integrate all the activities of the organs
(B) Coordination is the activities of the organs two or more organs interact and complement the functions of one another
(C) The functions of muscles, lungs, heart, blood vessels, kidney and other organs are coordinated while performing physical exercises
(D) Short fibres which branch repeatedly and project out of the cell body also contain nissl's granules and are called dendrites

How many of the above satements are true :-

- (1) 4 (2) 3 (3) 1 (4) 2

119. Which of the following statement is/are true about resting phase of neuron ?

- (a) Voltage gated channels not work.
(b) $\text{Na}^+ - \text{K}^+$ pump works for outward of 3K^+ ions and inward of 2Na^+ ions
(c) Move premiable for Na^+
(d) Potential differece across the membrane at rest is called resting potential
(1) b, c, d only
(2) a, c, d only
(3) a & d only
(4) a, b, c, d

120. Nerve fibres are myelinated or unmyelinated. What is true about unmyelinated nerve fibres in PNS.

- (1) Covered with oligodendrocytes
(2) Myelin sheath is present
(3) Layer of schwann cells is absent
(4) Layer of schwann cells present but that do not from myelin sheath

121. Match these columns :-

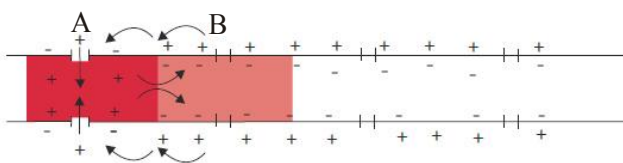
Column-I		Column-II	
A.	CNS	(i)	Cranical and spinal nerves
B.	PNS	(ii)	Relay impulses from CNS to skeletal muscles
C.	Somatic neural system	(iii)	Transmits impulses from CNS to involuntary organs and smooth muscles
D.	Autonomic neural system	(iv)	Site of information processing and control

- | A | B | C | D |
|--------|-----|-----|-----|
| (1) iv | iii | i | ii |
| (2) iv | i | iii | ii |
| (3) iv | i | ii | iii |
| (4) iv | ii | iii | i |

122. The type of fibres which carry nerve impulses from central nervous system to target organs or neuromuscular junction are known as :-

- (1) sensory fibres (2) afferent fibres
(3) muscle fibres (4) efferent fibres

123. How many given points are correct for following diagram at site A :-



- (i) $\text{Na}^+ - \text{K}^+$ pump is open
(ii) Outer surface is -ve charged due to K^+ influx
(iii) K^+ VGC is closed
(iv) Na^+ VGC are open
(v) -60 mv voltage
(vi) Passive channels open
(vii) Nerve impulse develops
(viii) Membrane is more permeable for K^+
- (1) Three (2) Six
(3) Four (4) Five

124. Lamprey is not a fish since it does not have

- (1) Closed blood vascular system
(2) Fins
(3) Body scales
(4) Cranium and vertebral column

125. Sea horse is

- (1) A marine mammal
(2) A marine fish
(3) A cantilegious fish
(4) A lung fish

126. Which of the following is the exclusive feature of the birds

- (1) 4-chambered heart
(2) Homoeothermic
(3) Fore limbs modified into flippers
(4) Presence of feathers

127. Mark the correct statement for the phyla, Platyhelminthes to Echinodermata.

- (1) All groups represent organ-system level of organization
(2) All are triploblastic
(3) All are bilaterally symmetrical
(4) All are parasite

128. When any plane passing through the central axis of the body divides the organism into two identical halves, the organism is called

- (1) Radially symmetrical
(2) Bilaterally symmetrical
(3) Asymmetrical
(4) Metamerically segmented

129. In Platyhelminthes

- (1) The embryonic layers, ectoderm and endoderm, are separated by mesoglea
(2) The body is asymmetrical
(3) There is tissue level of organization
(4) The body cavity is absent

130. Which of the following is not a chordate character

- (1) Presence of paired pharyngeal gill slits
- (2) Ventral heart
- (3) Solid and ventral nerve cord
- (4) Presence of post anal tail

131. Sycon belongs to a group of animals which are best described as

- (1) Unicellular or acellular
- (2) Multicellular without any tissue organization
- (3) Multicellular with a gastrovascular system
- (4) Multicellular having tissue organization but no body cavity.

132. Which one of the following statements about all the four of **Spongilla**, Leech, Dolphin and Penguin is correct ?

- (1) All are bilaterally symmetrical
- (2) Penguin is homoiothermic while all others are Poekilotheurms
- (3) Leech is a fresh water form while all others are marine
- (4) Spongilla has special collared cells called choanocytes, not found in the remaining three

133. Match item in column A with those given in column B.

Column A	Column B
A. Limbless reptile	1. Lamprey
B. Jawless vertebrate	2. Salamander
C. Amphibian	3. Snake
D. Cartilaginous fish	4. Shark
E. Flightless bird	5. Ostrich

- (1) A-1, B-2, C-3, D-4, E-5
- (2) A-2, B-1, C-3, D-4, E-5
- (3) A-3, B-1, C-2, D-4, E-5
- (4) A-4, B-2, C-3, D-1, E-5

134. Which one of the following statements is totally wrong about the occurrence of notochord while the other three are correct

- (1) It is absent throughout life in humans from the very beginning
- (2) It is present throughout life in *Amphioxus*
- (3) It is present only in larval tail in *Ascidians*
- (4) It is replaced by a vertebral column in adult frog

135. Bath sponge is common name of -

- (1) *Spongilla*
- (2) *Euspongia*
- (3) *Euplectella*
- (4) *Leucosolenia*

136. Which of the following character does not belong to Arthropoda :-

- (1) Closed circulatory system
- (2) Movable jaws
- (3) Segmented body
- (4) Jointed legs

137. The canal system is characteristic feature of -

- (1) sponges
- (2) helminthes
- (3) echinoderms
- (4) coelenterates

138. Sea-anemone belongs to phylum

- (1) Protozoa
- (2) Porifera
- (3) Coelenterata
- (4) Echinodermata

139. Head of cockroach according to its position is known as :

- (1) Hypopharynx
- (2) Hypocerebral
- (3) Hypognathus
- (4) Supragnathus

140. Number of segments in cockroach leg :

- (1) 3
- (2) 5
- (3) 6
- (4) 9

141. The dorsal plate of skeleton found on the abdomen of cockroach is called :

- (1) Pleuron
- (2) Sternum
- (3) Tergum
- (4) Vertex

142. Anal styles are found in :

- (1) Housefly
- (2) Female cockroach
- (3) Male cockroach
- (4) Both male & female cockroach

- 143.** Exoskeleton of cockroach is made up of :
 (1) Cartilage (2) Cuticle
 (3) Chitin (4) Amino acids
- 144.** Nissl's bodies found in neurons are :-
 (1) Made of DNA
 (2) Masses of ribosome and RER
 (3) Help in formation of neurofibrils
 (4) Masses of mitochondria
- 145.** Afferent nerve fiber conducts impulse from :-
 (1) C.N.S. to effector
 (2) Receptor to C.N.S.
 (3) Receptor to effector
 (4) Effector to receptor
- 146.** The nerves leading to the central nervous system are called :-
 (1) Afferent (2) Efferent
 (3) Motor (4) None
- 147.** Integrative system in the body are :-
 (1) Endocrine system (2) Nervous system
 (3) Blood vascular system (4) Both 1 & 2
- 148.** Rapid integration of the functional activities in human is achieved by :-
 (1) Nervous system
 (2) Endocrine system
 (3) Blood
 (4) Muscular system
- 149.** Sheath of schwann occurs on :-
 (1) Neurons (2) Axons
 (3) Dendrons (4) Neuroglia
- 150.** Nerve fibres are surrounded by an insulating fatty layer called :-
 (1) Adipose sheath (2) Myelin sheath
 (3) Hyaline sheath (4) Peritoneum
- 151.** Which of the following statement is correct for node of Ranvier of nerve ?
 (1) Neurilemma is discontinuous
 (2) Myelin sheath is discontinuous
 (3) Both neurilemma & Myelin sheath are discontinuous
 (4) Covered by myelin sheath
- 152.** Non Myelinated axons differ from myelinated in that they :-
 (1) Are more excitable
 (2) Lack nodes of Ranvier
 (3) Are not capable of regeneration
 (4) Are not associated with Schwann cells
- 153.** Myelin sheath covers which of the following ?
 (1) Muscle fibre (2) Nerve fibre
 (3) Collagen fiber (4) Tendons
- 154.** The gray matter differs from white matter in the :-
 (1) Absence of axons
 (2) Absence of myelin sheath
 (3) Presence of myelin sheath
 (4) Absence of neurilemma
- 155.** Function of nervous tissue is :-
 (1) Irritability or Excitability (2) Sensitivity
 (3) Elasticity (4) Contraction
- 156.** Which cell in our body is more than a feet long ?
 (1) Nerve cell (2) Muscle cell
 (3) Bone cell (4) Gland cell
- 157.** Saltatory conduction occurs in :-
 (1) Non-myelinated fibers
 (2) Myelinated fibers
 (3) Both of them
 (4) None of them
- 158.** Nerve impulses are initiated by nerve fibers only when the membrane shall become more permeable to :-
 (1) Adrenaline (2) Phosphorus
 (3) Sodium ions (4) Potassium ions
- 159.** A short period during which a nerve is unable to conduct nerve impulse is called :-
 (1) Synaptic delay
 (2) Refractory period
 (3) Resting potential
 (4) Critical period

- 160.** Which of the following is dominant intracellular anion :-
 (1) Potassium (2) Chloride
 (3) Phosphate (4) Calcium
- 161.** Acetylcholinesterase help in breaking :
 (1) synapse (2) acetylcholine
 (3) dendrites (4) axon
- 162.** Corpus callosum connects :-
 (1) Two cerebral hemisphere
 (2) Two optic lobes
 (3) Two olfactory lobes
 (4) Optic chiasma
- 163.** The membrane which cover the brain and the spinal cord is :-
 (1) White matter (2) Grey matter
 (3) Peritonium (4) Menix
- 164.** The box like bony structure which encloses the brain is called :-
 (1) Cranium (2) Pericardium
 (3) Peritoneum (4) Periosteum
- 165.** Human brain is covered by a hard layer called -
 (1) White matter (2) Duramater
 (3) Piamater (4) Gray matter
- 166.** Which part of cerebrum convert short term memory into long term memory –
 (1) Limbic lobe
 (2) Temporal lobe
 (3) Prefrontal cortex
 (4) Hippocampal lobe
- 167.** Respiratory control in brain occurs in :-
 (1) Medulla oblongata (2) Cerebellum
 (3) Hypothalamus (4) Pericardium
- 168.** Which part of brain is supposed to be damaged if in an accident, a person lost control of water balance, hunger and body temp. :-
 (1) Cerebellum
 (2) Hypothalamus
 (3) Medula oblongata
 (4) Corpora quadrigemina
- 169.** Brain of human differs from that of frog in having:-
 (1) Large olfactory lobe
 (2) Small hypothalamus
 (3) Small cerebellum
 (4) Corpus callosum
- 170.** Ventral root of spinal nerve has :-
 (1) Sensory fibers
 (2) Motor fibers
 (3) Sensory and motor fibers both
 (4) None of these
- 171.** The "butter fly" like structure surrounding the central cavity of human's spinal cord is called :-
 (1) Funiculus (2) Horn
 (3) White matter (4) Gray matter
- 172.** Which cranial nerve is the longest and supplies all parts of body other than head :-
 (1) Trochlear nerve (2) Vagus nerve
 (3) Oculomotor nerve (4) Auditory nerve
- 173.** Number of spinal nerves in human :-
 (1) 31 pairs (2) 32 pairs
 (3) 12 pairs (4) 37 pairs
- 174.** In human, autonomic nervous system is composed of :-
 (1) Sympathetic and parasympathetic nerves
 (2) Cranial and spinal nerves
 (3) Brain and spinal nerves
 (4) Medullated and non-medullated nerves
- 175.** All spinal nerves are :-
 (1) Motor (2) Sensory
 (3) Mixed (4) None of the above
- 176.** Which of the following nerve helps in maintaining the equilibrium of body :-
 (1) Trochlear (2) Abducens
 (3) Auditory (4) Facial

177. Autonomic nervous system controls :-

- (1) Conditioned reflexes
- (2) Functioning of spinal cord
- (3) Functioning of visceral organs
- (4) Reflex actions

178. Which of the following spinal nerves does not found in human :-

- (1) Caudal nerves
- (2) Sacral nerves
- (3) Cervical nerves
- (4) Lumbar nerves

179. The cranial and spinal nerves are included under :-

- (1) Autonomic nervous system
- (2) Peripheral nervous system
- (3) Central nervous system
- (4) Cutaneous nervous system

180. Norepinephrine leads to increase in :-

- (1) Blood pressure
- (2) Urine production
- (3) Cellular respiration
- (4) Release of epinephrine